

DIAS GEORGE

Software Development Manager | AI Transformation

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EXPERIENCE SUMMARY

R&D MANAGER

APTEAN INDIA PVT LTD

2019 - PRESENT

- Spearheaded the development and continuous enhancement of GenomeQuest, delivering scalable solutions that support global patent and sequence search for the life sciences sector.
- Championed the adoption of AI-assisted engineering workflows, improving developer productivity and accelerating software delivery.

MANAGING DIRECTOR / WEB DEVELOPER

LIQUID SOUL DIMENSIONS LLP

2017 - 2018

- Founded and led a software/digital consultancy, developing and delivering 5 custom web applications .
- Managed advertising engagements for clients including GE and OnePlus.

BUSINESS ANALYST / TEAM LEAD

IBM INDIA PVT LTD

2007 - 2016

- Delivered 20+ projects and led a team maintaining 15 mission-critical AT&T applications across IBM's SDLC.
- Resolved 20+ Severity-1 incidents and 200+ support tickets, maintaining 96% uptime while cutting resolution time by 50%.

PROFILE

Software Development Manager with 18+ years of experience leading the design, development, and delivery of enterprise software solutions. I have progressed from hands-on software engineering into engineering leadership, building high-performing teams, defining technical strategy, and delivering scalable, business-critical applications.

I enjoy solving complex engineering challenges, mentoring engineers, and improving the way teams deliver software. More recently, I have led AI transformation initiatives that embed AI-assisted development practices to improve productivity, accelerate delivery, and enhance software quality, while maintaining a hands-on approach to technology and engineering leadership.

CAREER HIGHLIGHTS

- Experienced in defining product roadmaps, driving software architecture, leading Agile development teams and delivering scalable, business-critical applications.
- Reduced feature delivery time by 40% by introducing AI-assisted workflows (Claude, ChatGPT, Gemini, MCP) across a 9-person engineering team.
- Created customer features using AI models and MCP architecture to bring a chat-based customer workflow that replicates application functions.
- Led global engineering teams across India and the US, mentoring engineers, driving technical excellence and delivering high-impact product enhancements.
- Grew application release cadence from 2 to 4 releases per year while streamlining stakeholder approvals and release documentation, maintaining 95% customer retention throughout.
- Delivered enterprise software solutions used by leading pharmaceutical companies, biotechnology organisations, patent professionals and Fortune 500 clients.
- Extensive experience across the software development lifecycle, including product strategy, architecture, application development, testing, production support and engineering management.

CORE COMPETENCIES				
	Leadership & Management	Software Engineering	AI & Modern Engineering	Technical Program Delivery
	Engineering Leadership	Software Architecture	AI Transformation	Agile, Scrum & Kanban
	Team Building & Mentoring	Enterprise Application Development	AI-Assisted Development	Technical Program Delivery
	Cross-functional Leadership	System Design	Model Context Protocol (MCP)	Release Management
	Performance Management	Backend Engineering	AI Workflow Automation	SDLC Management
	Technical Strategy	API & Service Design	Claude, ChatGPT & Gemini	Product Roadmap Planning
	Project & Delivery Management	Application Modernization	AI Agent Development	Stakeholder Management
	Resource Planning	CI/CD & DevOps Practices	AI-enabled Engineering Workflows	Requirements Analysis
	Vendor & Budget Management	Database Engineering	AI Adoption & Enablement	GDPR & Data Compliance
EDUCATION				
COLLEGE OF ENGINEERING, TRIVANDRUM				
2003 - 2007				
Bachelor of Electrical and Electronics/ B.Tech				
7.5 GPA				
TECHNICAL SKILLS				
CLOUD & INFRASTRUCTURE				
Docker, Jenkins, Hetzner Cloud, Server Virtualization, Physical-to-VM Migration				
SOFTWARE ENGINEERING & ARCHITECTURE				
Enterprise Application Development, System Design, Software Architecture, API Design, SDLC				
DEVELOPMENT PRACTICES				
Agile, Scrum, Kanban, CI/CD, Code Reviews, Test Automation, Azure DevOps (Boards)				
PROGRAMMING LANGUAGES				
- Python, JavaScript, PHP, Java, PL/SQL, C, C++, Shell, Perl - Flutter, WebSockets, REST APIs, SOAP Web Services, HTML5, CSS3				
DATABASES & PLATFORMS				
PostgreSQL, MySQL, Oracle, DB2, Linux/UNIX, Windows Server, Git				
PROJECTS				
28 – MULTIPLAYER CARD GAME				
WWW.BID28.COM				
CRICKET STATS WEBSITE (DEMO)				
WWW.DIASGEORGE.COM/HOF-DEMO/LIVE.HTML				
LANGUAGES				
- English - Hindi - Malayalam				

DETAILED PROFESSIONAL EXPERIENCE

APTEAN INDIA PVT. LTD

R&D MANAGER	2025 - PRESENT
Promoted from Lead Engineer to Team Lead and subsequently R&D Manager, leading the engineering direction of the GenomeQuest platform, one of the industry's leading biological patent intelligence solutions used by pharmaceutical companies, biotechnology organisations and patent professionals. TECHNOLOGIES Python, PostgreSQL, Linux, AI (Claude, ChatGPT, Gemini), Model Context Protocol (MCP), Patent Data Processing, Agile	LEADERSHIP & RESPONSIBILITIES • Lead a global engineering team of 9 members (6 in India and 3 in the US). • Define the technical roadmap in collaboration with Product Management. • Drive architectural decisions, sprint planning and engineering best practices. • Conduct performance reviews, mentoring and career development for engineering team members. • Oversee backend engineering, patent content production and platform reliability. • Collaborate with Product Management, QA and UI/UX teams to deliver product releases.

KEY ACHIEVEMENTS

- Rolled out AI-assisted engineering workflows (Claude, ChatGPT, Gemini, MCP) across software and content development, cutting estimated development effort per feature by roughly 40% and establishing reusable AI-workflow standards adopted across both the India and US teams.
- Delivered the Chemical Modification Search module, extending platform search coverage to 10 new use cases and driving adoption by around 20 new enterprise clients.
- Designed and implemented an MCP-based AI platform allowing designers and developers to perform UI/UX tasks through intelligent agents.
- Supported a product used by more than 100 enterprise customers, including global pharmaceutical companies, biotechnology organisations and patent law firms.
- Owned vendor budgeting and selection for external data providers, building multi-vendor data ingestion pipelines while evaluating AI model providers on cost-performance tradeoffs to select solutions fitting use case and budget constraints.
- Transitioned the team's delivery process from Scrum to Kanban to improve throughput and productivity.
- Increased application release cadence from 2 to 4 releases per year by streamlining stakeholder approvals, release documentation and customer release notes, while maintaining 95% customer retention.
- Oversaw migration of physical servers to a virtualized infrastructure environment, improving resource utilization and deployment flexibility.
- Containerized and deployed MCP server infrastructure using Docker on Linux, supporting the team's AI-assisted development workflows.
- Own CI/CD pipeline delivery for the GenomeQuest platform using Jenkins, supporting automated builds and the increase in release cadence from 2 to 4 releases per year.
- Own end-to-end Agile delivery cadence for the team, including stakeholder board management, sprint planning, reviews and retrospectives.

DETAILED PROFESSIONAL EXPERIENCE

APTEAN INDIA PVT. LTD

TEAM LEAD

2021 - 2025

Promoted to Team Lead to establish and grow the India engineering team responsible for backend engineering and content production for GenomeQuest.

TECHNOLOGIES

Python, PostgreSQL, Linux, Patent Data Processing, Agile

RESPONSIBILITIES

- Built and managed the India content engineering team.
- Planned sprint deliveries and coordinated releases with global engineering teams.
- Mentored engineers and established development standards.
- Improved production monitoring and engineering processes.
- Worked closely with Product Management on feature planning.

KEY ACHIEVEMENTS

- Reduced patent publication turnaround times through improvements to content production workflows.
- Improved patent family processing and production stability.
- Automated several manual engineering and production activities.
- Successfully onboarded and mentored new engineering team members.

LEAD ENGINEER

2019 - 2021

Joined the GenomeQuest engineering team responsible for backend development and patent content production.

TECHNOLOGIES

Python, PostgreSQL, PERL, JAVA, Linux

RESPONSIBILITIES

- Backend application development.
- Patent ingestion and processing pipelines.
- Production support and issue resolution.
- Process automation.
- Content validation and database updates.

KEY ACHIEVEMENTS

- Automated biological sequence database build processes.
- Improved PATSTAT patent family processing.
- Enhanced public database build processes, including UNIPROT and GenBank.
- Improved production build reliability and reduced manual intervention.
- Worked with external content providers to streamline data integration and production workflows.

DETAILED PROFESSIONAL EXPERIENCE

LIQUID SOUL DIMENSIONS LLP

MANAGING DIRECTOR /SENIOR WEB DEVELOPER

2017 - 2018

Founded and managed a software and digital solutions company delivering custom web applications and digital media solutions.

TECHNOLOGIES

PHP, Laravel, JavaScript, HTML5, CSS3, MySQL

RESPONSIBILITIES

- Business management and client engagement.
- Solution architecture and software design.
- Full-stack web development.
- Project planning and delivery.
- Technical consulting

KEY ACHIEVEMENTS

- Successfully delivered web development projects including the IDEC 2018 conference platform and the company's corporate site, alongside creative and advertising engagements for brands including GE and OnePlus.
- Managed complete project lifecycles from requirements gathering through deployment.
- Delivered projects within committed timelines while maintaining strong client relationships.

IBM INDIA PVT LTD.

TEAM LEAD

2015 - 2016

Progressed into a Team Lead role responsible for development, maintenance and production support of 15 AT&T enterprise applications supporting Provisioning, Inventory and Capacity Management.

TECHNOLOGIES

C, C++, Java, Oracle, Informix, Sybase, Shell, Perl, SOAP, Artix, Tuxedo, CORBA

RESPONSIBILITIES

- Led a team of five software engineers.
- Managed project planning, technical design and delivery.
- Coordinated production support and release activities.
- Conducted code reviews and mentored team members.
- Worked directly with AT&T stakeholders on project delivery.

KEY ACHIEVEMENTS

- Delivered more than 20 enterprise development projects.
- Managed over 200 production support requests and resolved more than 20 Severity 1 incidents.
- Became Subject Matter Expert across more than 30 enterprise applications.
- Successfully led multiple application modernization initiatives.

DETAILED PROFESSIONAL EXPERIENCE

IBM INDIA PVT LTD.

BUSINESS ANALYST / APPLICATION PROGRAMMER

2011 - 2015

Progressed from Application Programmer to Business Analyst, taking ownership of requirements, design, development, testing and production support for enterprise applications.

TECHNOLOGIES

C, C++, Oracle, DB2, Sybase, Shell, Perl, HP-UX, Solaris

RESPONSIBILITIES

- Requirements gathering and technical design.
- Application development.
- Unit, integration and system testing.
- Production support.
- Documentation and release management.

KEY ACHIEVEMENTS

- Delivered more than 20 development projects across the complete SDLC.
- Supported over 15 mission-critical AT&T applications.
- Resolved complex production issues with minimal business impact.
- Frequently assigned to high-priority projects requiring rapid technical problem solving.

ASSOCIATE SYSTEMS ENGINEER

2007 - 2011

Completed IBM technical training before joining the AT&T account as a software engineer supporting enterprise telecommunications applications.

TECHNOLOGIES

UNIX, C, C++, Oracle, DB2, Sybase, Perl, Shell

RESPONSIBILITIES

- Software development using C/C++.
- Unit, system and interface testing.
- Production support.
- Application maintenance.
- Technical documentation.

KEY PROJECTS

- Connect:Direct interface implementation.
- Provisioning application enhancements.
- Shell scripting automation.
- Backend development for Bell Labs applications.
- HP-UX and Solaris production support.

PERSONAL PROJECTS

MULTIPLAYER CARD GAME

2025

Designed and built a real-time multiplayer platform for "28" (Twenty-Eight), an Indian trick-taking card game popular in Kerala played by 4 players in fixed partnerships. Personally built the game engine, service architecture, and database schema, then directed AI coding agents to complete implementation across a 35,000+ line, three-service production codebase.

TECHNOLOGIES

- Client: Flutter 3.x (Dart), Riverpod, Freezed, Socket.IO client
- Backend: Python/FastAPI/SQLAlchemy/Alembic (API) + Node.js/Express/Socket.IO/Sequelize (game engine), PostgreSQL
- AI Bots: Node.js, Socket.IO client, Axios
- Infrastructure: Hetzner Cloud, systemd, Cloudflare, custom bash deploy pipeline (git + SSH + rsync)
- Built a real-time game engine in Node.js (Express + Socket.IO) driving the full two-round bidding phase, trump selection, and trick-play logic over WebSockets, synchronizing live 4-player sessions.
- Built a Python (FastAPI + SQLAlchemy/Alembic) REST API for auth, accounts, and scoring, and a cross-platform Flutter client (Riverpod + Freezed) shipping to Android, iOS, web, and desktop — both sharing one PostgreSQL database and JWT auth.
- Added rule-based AI bots that autonomously bid, select trump, and play tricks, and hardened the system for production with transaction-wrapped game mutations, bidding-phase/card-ownership gates, and bcrypt password hashing.
- Deployed and hosted on Hetzner Cloud (systemd services) behind Cloudflare, with a custom deploy script automating git sync, dependency installs, DB migrations, service restarts, and Flutter web builds.

CRICKET STATS WEBSITE(DEMO)

2026

Designed and built a full-stack cricket data and live-scoring platform covering 19 IPL seasons, 800+ players, live match scores, and fan voting. Personally designed the PostgreSQL schema, data pipelines, and core ranking algorithm, then directed AI coding agents to complete implementation across a 16,000+ line codebase spanning data pipelines, API, and frontend.

TECHNOLOGIES

- Client: Vanilla HTML/CSS/JS, Tailwind CSS, Chart.js, Firebase JS SDK
- Backend: Python, FastAPI, pycpg 3, PostgreSQL, Redis
- Auth: Firebase Admin SDK
- Infrastructure: Docker, docker-compose, nginx
- Built three data pipelines normalizing 19 seasons of squads/stats into PostgreSQL (with cross-source player name resolution), and a Python/FastAPI REST API (~20 endpoints) with Firebase auth (Google + phone OTP) gating a one-vote-per-user ballot system.
- Integrated a live-scoring provider with Redis caching, shipped a responsive HTML/JS/Tailwind frontend with leaderboards and live match modals, and deployed via Docker (Postgres, Redis, FastAPI, nginx), validated with load tests.